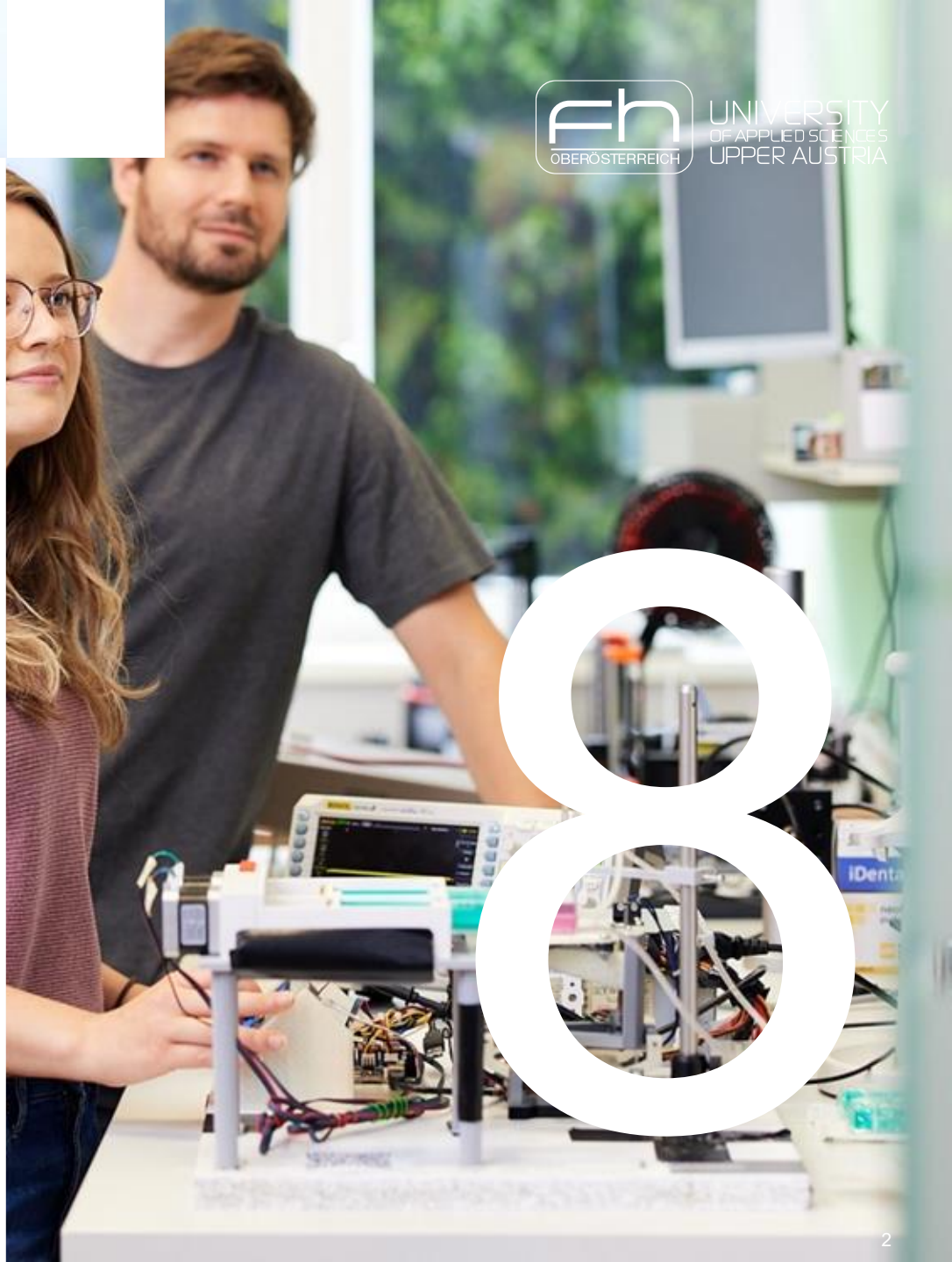
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UNIVERSITY OF APPLIED SCIENCES UPPER AUSTRIA

Game Engine Architectures

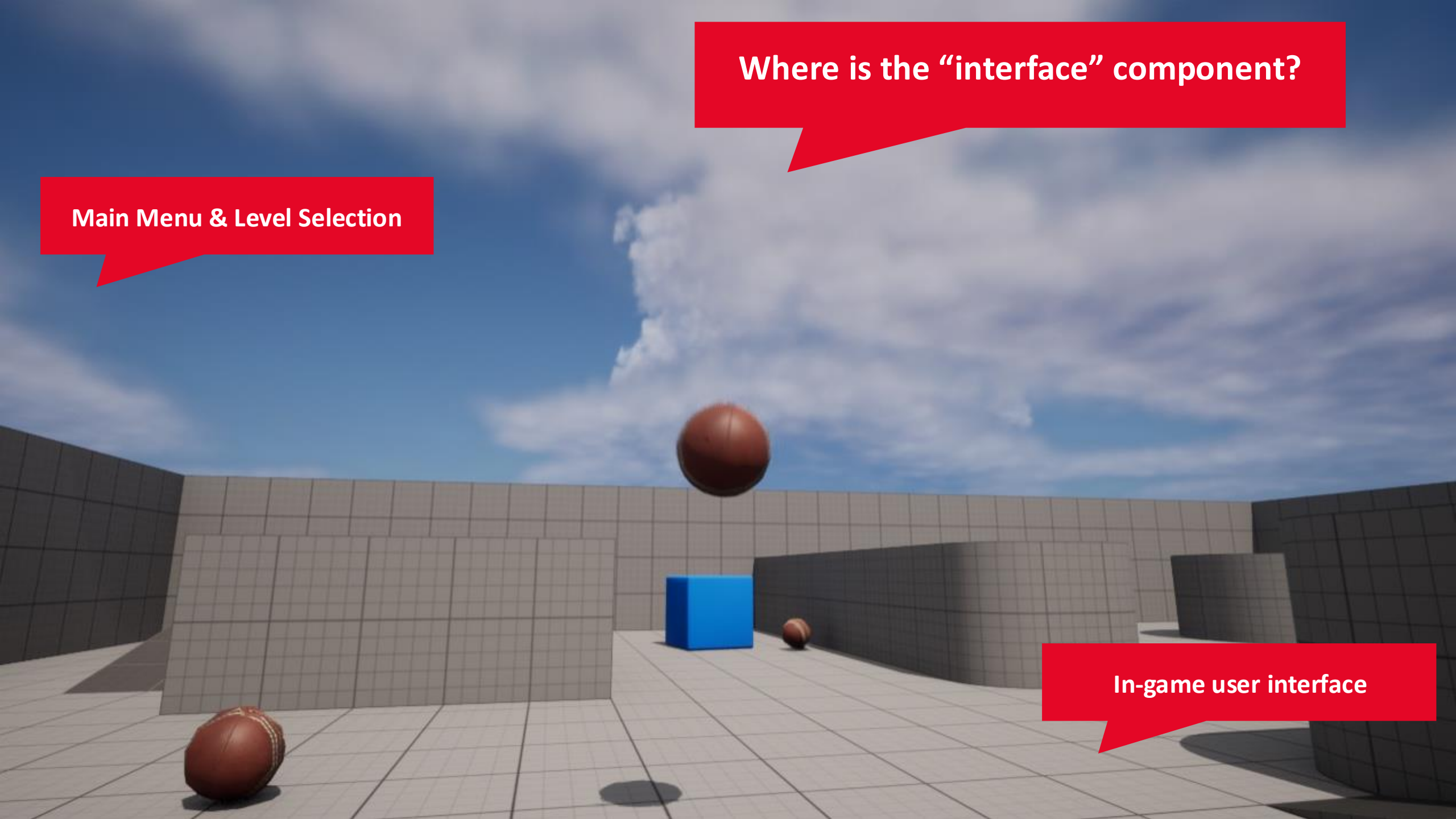
Fachhochschule Oberösterreich

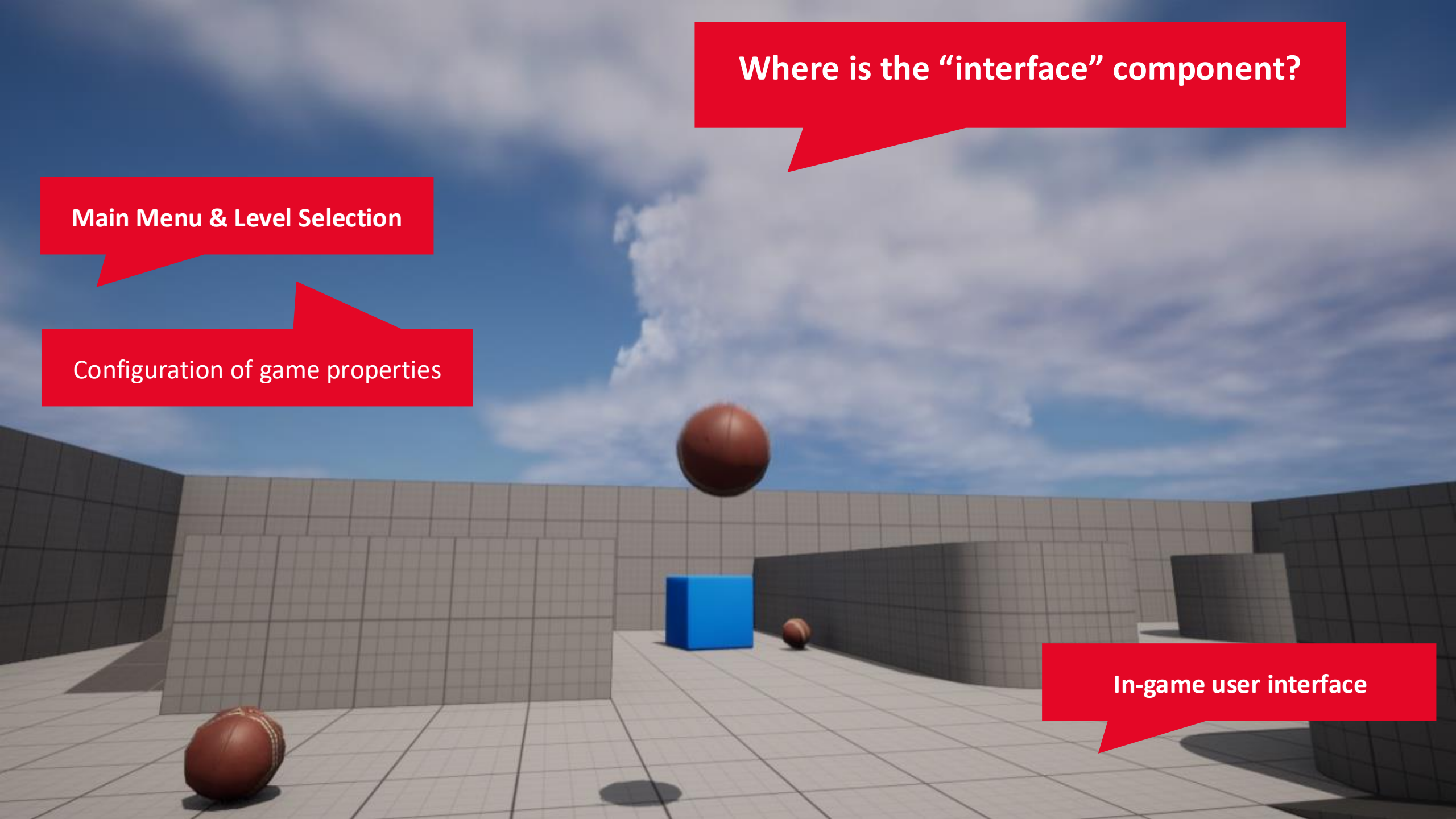
Game Evaluation Definition and Methods

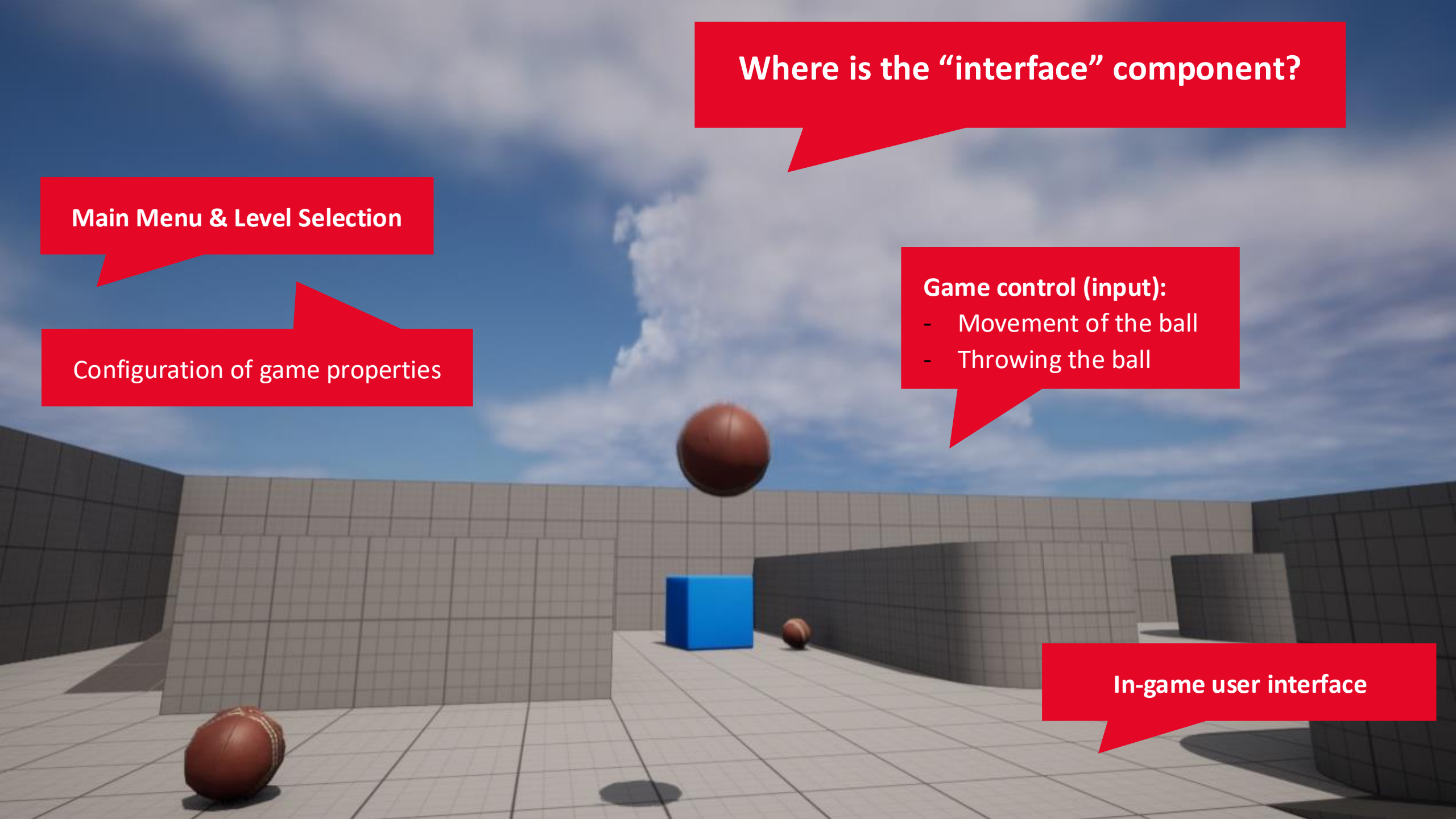


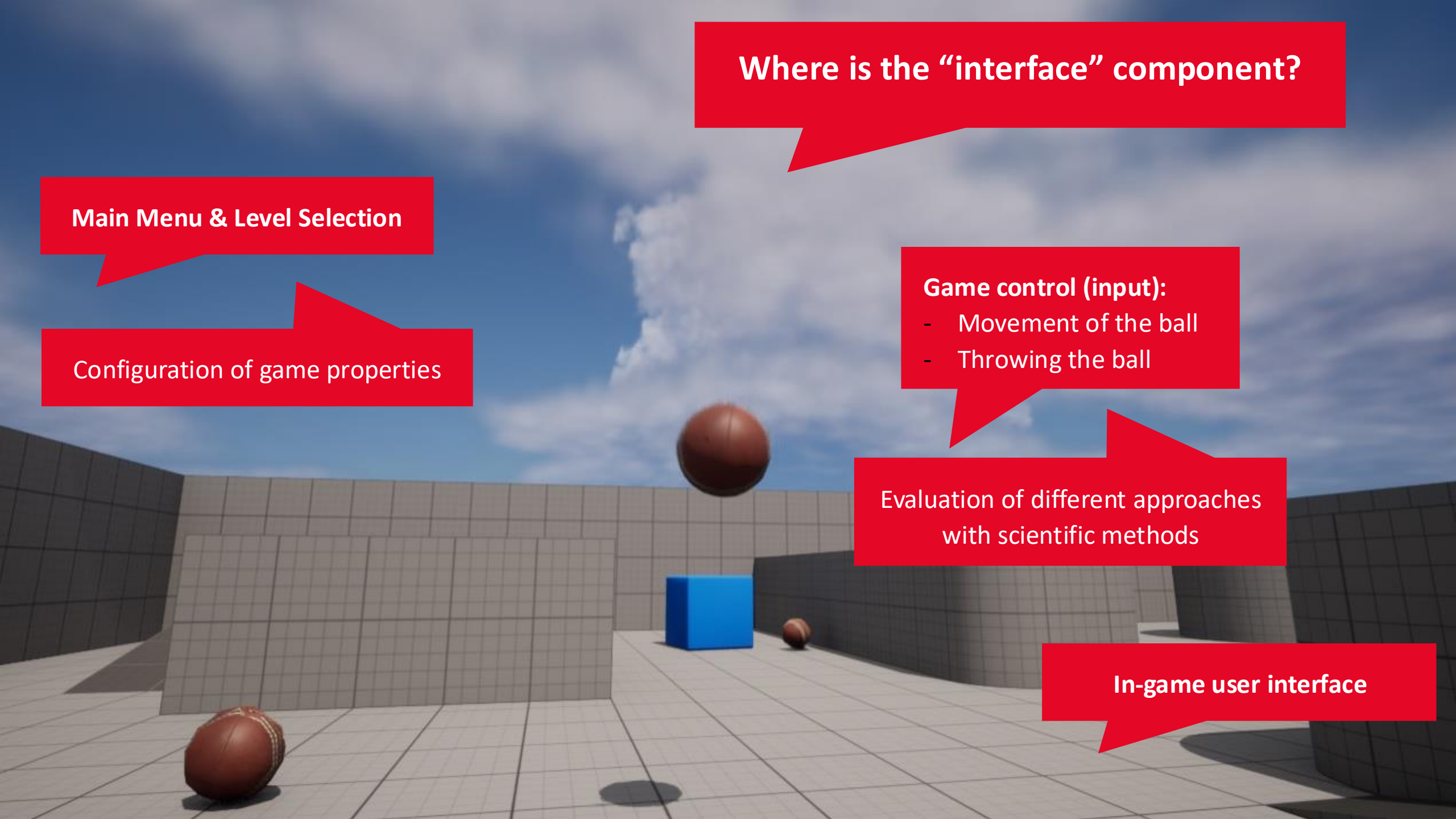

Where is the “interface” component?

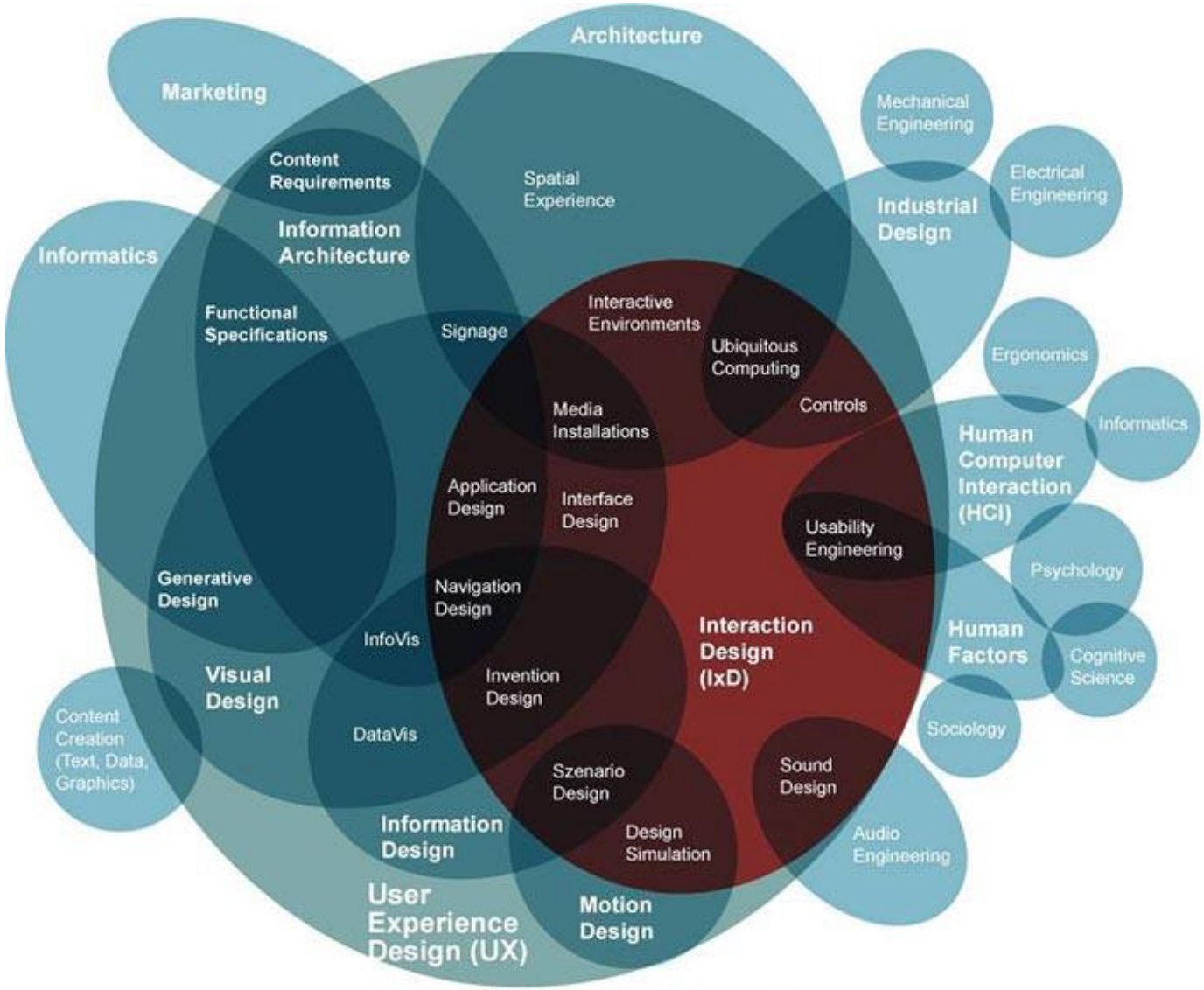

Where is the “interface” component?
Main Menu & Level Selection


Where is the “interface” component?
Main Menu & Level Selection
In-game user interface


Where is the “interface” component?
Main Menu & Level Selection
Configuration of game properties
In-game user interface


Where is the “interface” component?
Main Menu & Level Selection
Configuration of game properties
Game control (input):
- Movement of the ball
- Throwing the ball
In-game user interface


Where is the “interface” component?
Main Menu & Level Selection
Configuration of game properties
Game control (input):
- Movement of the ball
- Throwing the ball
Evaluation of different approaches with scientific methods
In-game user interface



Copyright :envis precisely (2009)
based on »The Disciplines of User Experience« by Dan Saffer (2008)
www.kickerstudio.com/blog/2008/12/the-disciplines-of-user-experience

Human-Computer Interaction

Definitions, Constructs, & Key Terms

- “HCI is a discipline concerned with **the design, evaluation, and implementation of interactive computing systems** for human use and with the study of **major phenomena surrounding them**.” (Definition SIGCHI)
- “Human computer interaction is the **study of people, computer technology**, and the way these **influence each other**. We study HCI to determine how we can make computer **technology more usable by people**.” (Alan Dix, 2003)
- “Human-Computer Interaction (HCI) is the **study of interaction between people (users) and computers**. It is an interdisciplinary subjects, relating computer science with many other fields of study and research”
  - Cognitive science, psychology (theories of cognitive processes and analysis of user behavior)
  - Sociology and anthropology (interactions between technology, work, and organization)
  - Visual-, industrial-, interaction-, UX-design (creating interactive products)

[1] Dix et al (2003). Human computer interaction, 3rd Edition, Prentice Hall, ISBN 0130461091
[2] http://old.sigchi.org/cdg/cdg2.html
[3] http://en.wikipedia.org/wiki/Human-Computer_Interaction

Human-Computer Interaction

Definitions, Constructs, & Key Terms

- As there are so many disciplines involved, some fields differ in their focus

- **Human Factors**
  - More machine-centered, integrate hard and software involved
  - Vehicles, manufacturing, aviation, physical control panels, military, etc.
  - Performance-oriented, working with professionals (aviation, etc.)

- **Ergonomics**
  - Making products “fit” to users
  - Furniture, workstations, seating, etc.
  - https://youtu.be/rGq7TVOwA60

Human-Computer Interaction

Definitions, Constructs, & Key Terms

- **User Experience Design**
  - Research/design/engineering discipline dealing with interactions between humans and (digital) products

- **Human Centered Design, User Centered Design**
  - Structured processes aiming to elaborate “good” interfaces/interaction concepts (mostly used in the context of human-computer and human-machine interaction)

- **Interaction Design**
  - Allocating functions to humans or technology
  - Involves sequence/structure of touchpoints over space and time
  - It is not only about efficiency, but also about user experience as a whole

Benyon, D. (2019). Designing User Experience: a guide to HCI, UX and interaction design. Pearson UK.

Human-Computer Interaction

Automation

- **“The execution by a machine agent of a function that was previously carried out by a human”**

- Human history is a story of automation

- Example: Communicating a message to a distant receiver
  - Philippides: runs 42km from Marathon to Athens (490 BC)
  - You: text your peers, anywhere, anytime (today)
  - Between: An evolution of humans automating tasks
  - Printing press, trains, postal service, telegram, computers
  - Mobile computing: automating the task of “accessing a transmission device”



R. Parasuraman and V. Riley, Humans and automation: Use, misuse, disuse, abuse, Human factors, vol. 39, no. 2, pp. 230253, 1997.

Human-Machine Interaction

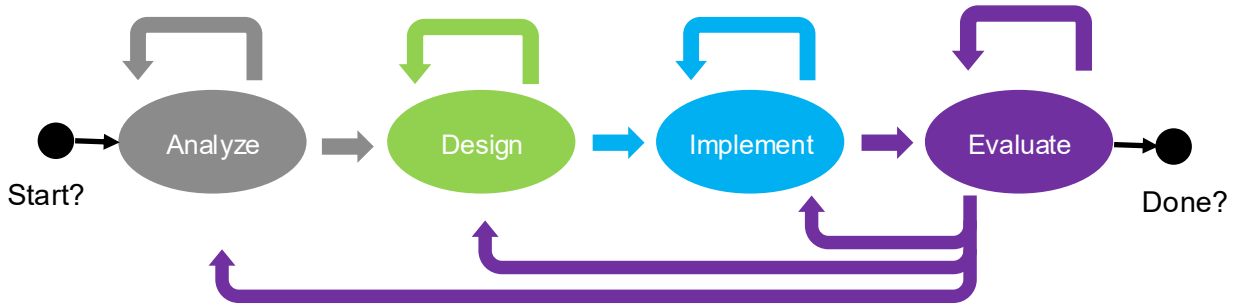
Human AND Machine vs. Human OR Machine

- **The “Levels of Automation” – The computer…**
  - 10 …decides everything, acts autonomously, ignoring the human
  - 09 ...informs the human only if it, the computer, decides to do
  - 08 …informs the human only if asked, or
  - 07 …executes automatically, then necessarily informs the human, and
  - 06 …allows the human a restricted time to veto before automatic execution, or
  - 05 …executes that suggestion if the human approves, or
  - 04 …suggests one alternative
  - 03 …narrows the selection down to a few, or
  - 02 …offers a complete set of decision/action alternatives, or
  - 01 …offers no assistance – human must take all decisions and actions

Parasuraman, R., Sheridan, T. B., & Wickens, C. D. (2000). A model for types and levels of human interaction with automation. IEEE Transactions on systems, man, and cybernetics-Part A: Systems and Humans, 30(3), 286-297.

Human-Centered Design

HCD = Design & Evaluate



Human-Centered Design

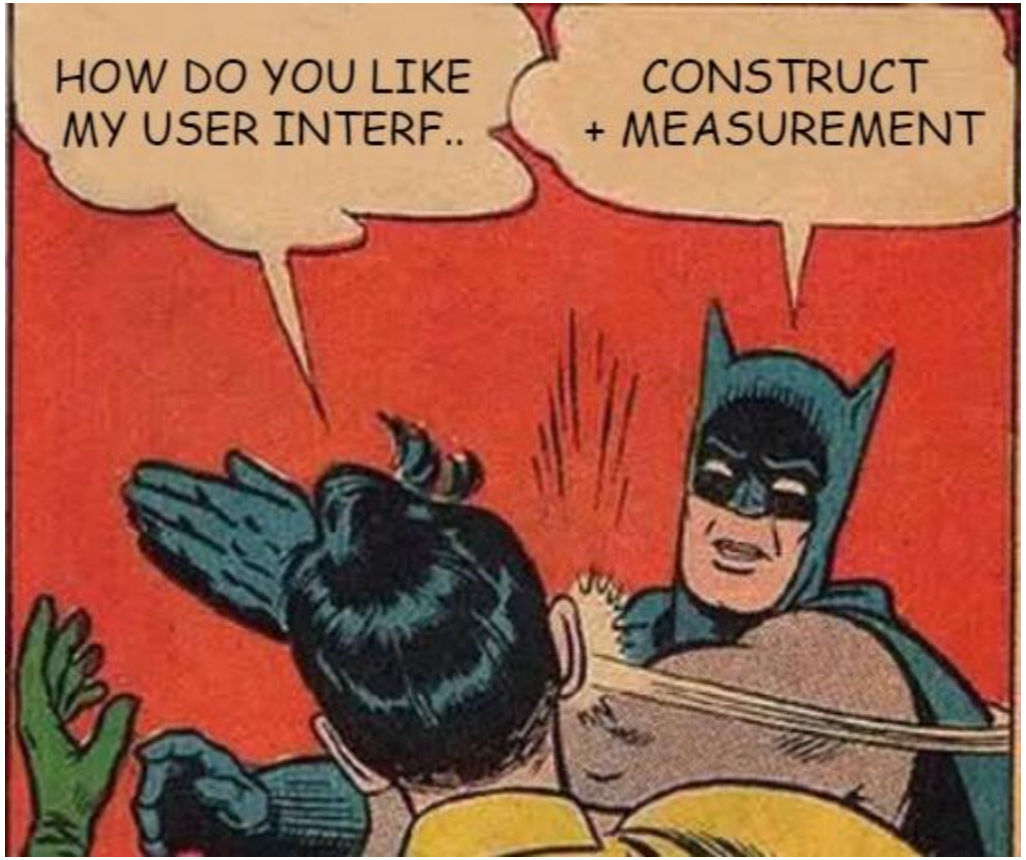
User Studies in a Nutshell

- **To evaluate a design/product we conduct an experiment to**
  - Evaluate the benefit of the solution, compared to a “baseline” (i.e., without solution)
  - Evaluate the effect of different designs, compared to other designs/products

- **Key Terms**
  - **Independent variable (IV):** the experimental condition
    - Design variant: Design 1 vs. Design 2 vs. Design 3, etc.
    - Other IVs are possible: Age (young vs. old), gender (female vs. male vs. diverse), etc.
  - **Dependent variable (DV):** the “thing” we are measuring
    - Quantitative: expressed in numbers (for example, questionnaire results)
    - Qualitative: expressed in words (for example, interview statements)

User Studies

What “thing” are we measuring?



User Studies

Constructs and Measurements

- **How should we evaluate?**
  - Imagine you have to **design a new experience for a Rollercoaster** in a Theme Park
  - Imagine you want to **design a new luxurious product** (say a watch)
  - Imagine you want to **design a new interface for a navigation system** in cars
  - Imagine you want to **develop a new and yet unknown games concept**

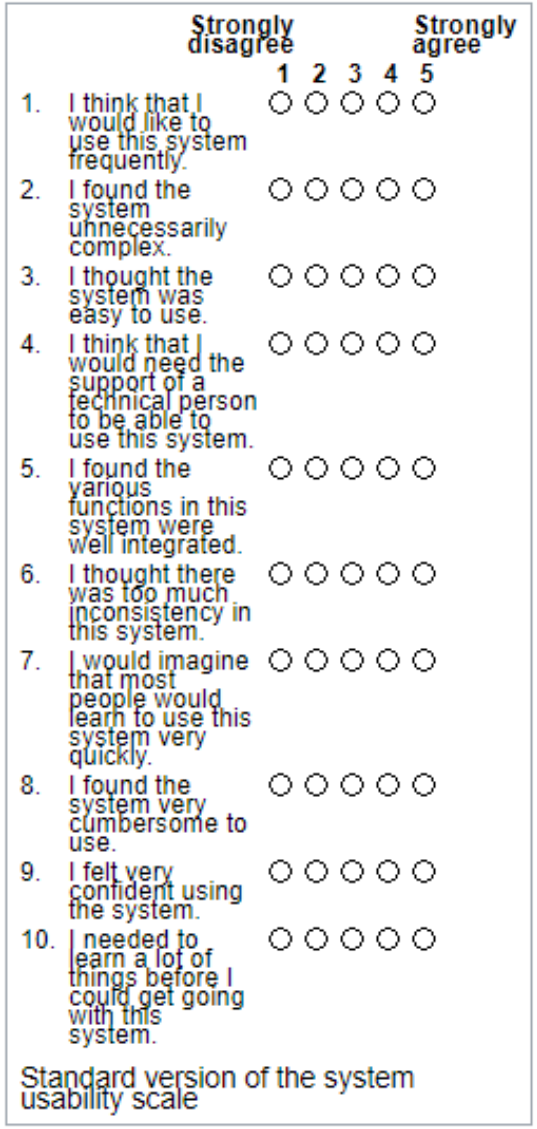
- **Knowledge and application of psychological constructs helps to understand your users, can help to determine if your requirements are satisfied, and allows to quantify different aspects of systems/products!**

User Studies

Usability (ISO 9241-210)

- **“Extent to which a system, product, or service can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use“**
  - Effectiveness: accuracy and completeness with which users achieve specified goals.
  - Efficiency: resources used in relation to the results achieved
  - Satisfaction: extent to which the user’s physical, cognitive, and emotional responses that result from the use of a system, product, or service meet the user’s needs and expectations

- **System Usability Scale**
  - 10 item scale, each rated from 1 to 5 (Likert scale)
  - Odd numbered items: score-1
  - Even numbered items: 5-score
  - Sum all re-calculated scores and multiply by 2.5
  - SUS ratings >80.3 (good), >68 (okay), <51 (bad)



		Strongly disagree				Strongly agree
		1	2	3	4	5
1.	I think that I would like to use this system frequently.	○	○	○	○	○
2.	I found the system unnecessarily complex.	○	○	○	○	○
3.	I thought the system was easy to use.	○	○	○	○	○
4.	I think that I would need the support of a technical person to be able to use this system.	○	○	○	○	○
5.	I found the various functions in this system were well integrated.	○	○	○	○	○
6.	I thought there was too much inconsistency in this system.	○	○	○	○	○
7.	I would imagine that most people would learn to use this system very quickly.	○	○	○	○	○
8.	I found the system very cumbersome to use.	○	○	○	○	○
9.	I felt very confident using the system.	○	○	○	○	○
10.	I needed to learn a lot of things before I could get going with this system.	○	○	○	○	○

Standard version of the system usability scale

User Studies

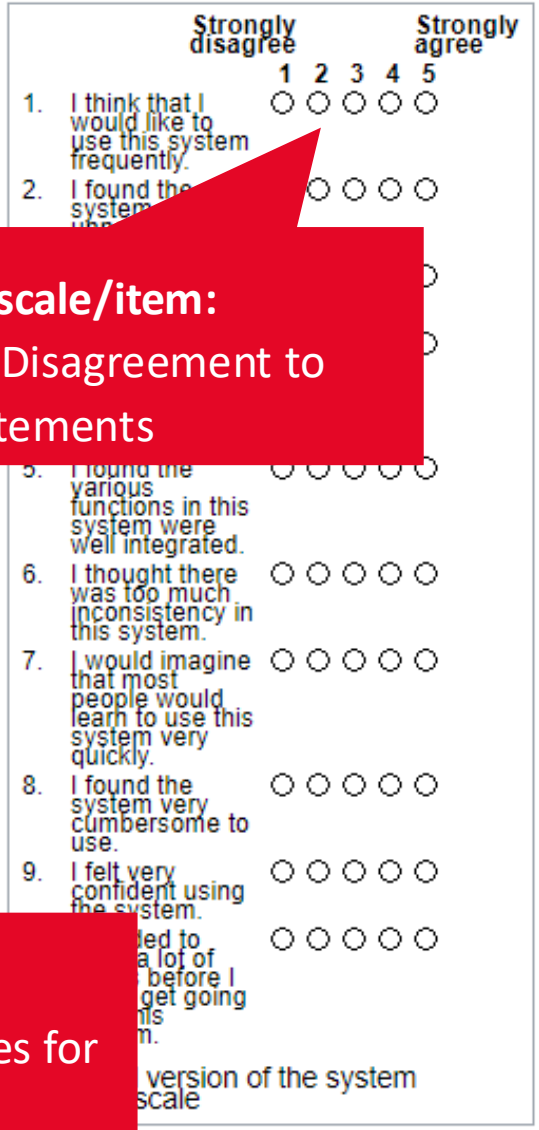
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  - Sum all re-calculated scores and multiply by 2.5
  - SUS ratings >80.3 (good), >68 (okay), <51 (bad)

Construct: "Usability"

Dimensions: Effectiveness, Efficiency, Satisfaction

Likert scale/item: Agreement/Disagreement to statements



		Strongly disagree 1	2	3	4	Strongly agree 5
1.	I think that I would like to use this system frequently.	○	○	○	○	○
2.	I found the system…	○	○	○	○	○
5.	I found the various functions in this system were well integrated.	○	○	○	○	○
6.	I thought there was too much inconsistency in this system.	○	○	○	○	○
7.	I would imagine that most people would learn to use this system very quickly.	○	○	○	○	○
8.	I found the system very cumbersome to use.	○	○	○	○	○
9.	I felt very confident using the system.	○	○	○	○	○
	…ed to …a lot of … before I get going …is …m.	○	○	○	○	○

… version of the system …scale

Measurement: A subjective scale. Can provide a single value, or different values for different construct dimensions

User Studies

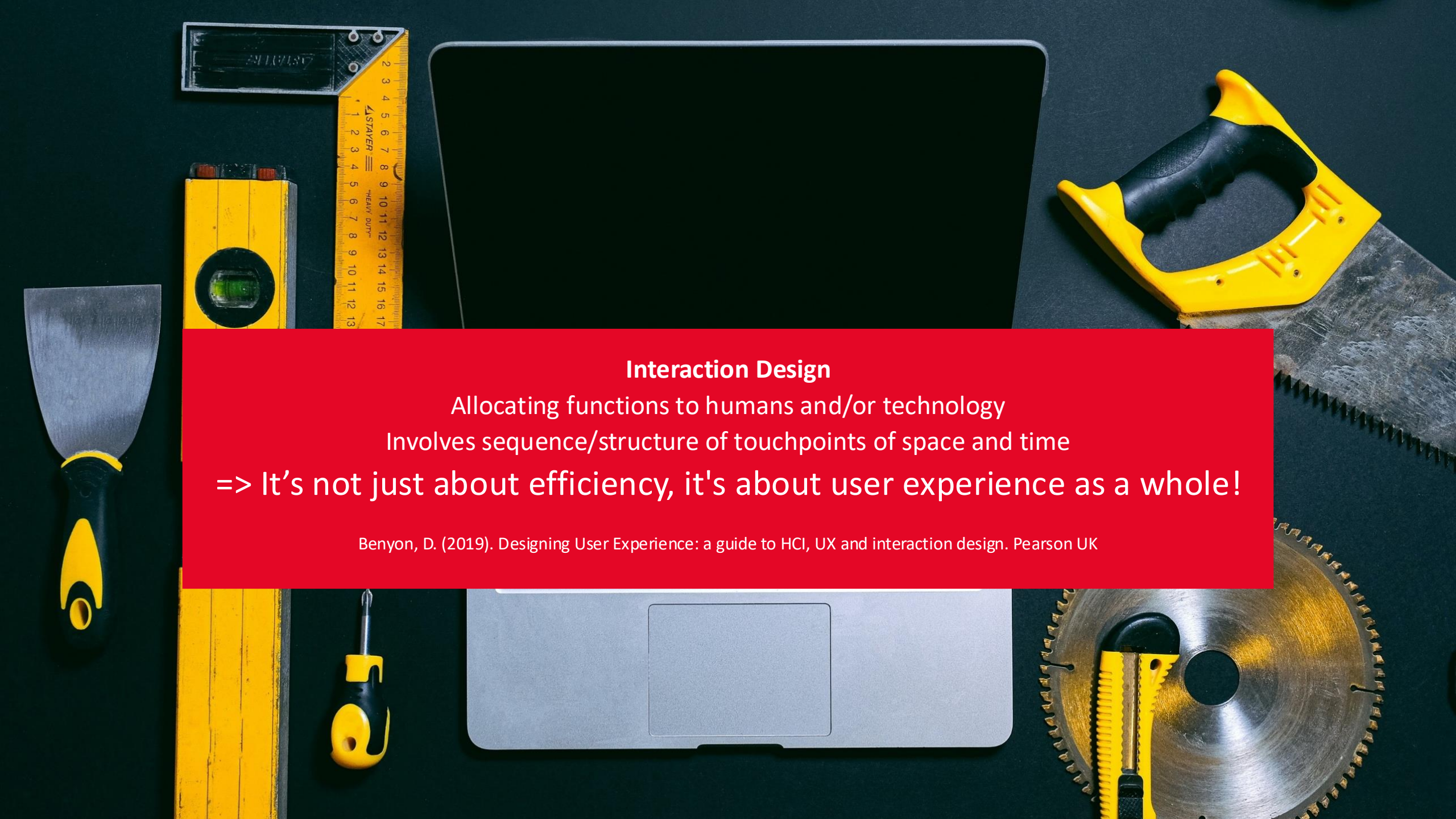
Game User Experience Satisfaction Scale (GUESS-18)

Game UX

Usability/Playability	I find the controls of the game to be straightforward.
	I find the game's interface to be easy to navigate.
Narratives	I am captivated by the game's story from the beginning.
	I enjoy the fantasy or story provided by the game.
Play Engrossment	I feel detached from the outside world while playing the game.
	I do not care to check events that are happening in the real world during the game.
Enjoyment	I think the game is fun.
	I feel bored while playing the game. (REVERSE CODE)
Creative Freedom	I feel the game allows me to be imaginative.
	I feel creative while playing the game.
Audio Aesthetics	I enjoy the sound effects in the game.
	I feel the game's audio (e.g., sound effects, music) enhances my gaming experience.
Personal Gratification	I am very focused on my own performance while playing the game.
	I want to do as well as possible during the game.
Social Connectivity	I find the game supports social interaction (e.g., chat) between players.
	I like to play this game with other players.
Visual Aesthetics	I enjoy the game's graphics.
	I think the game is visually appealing.

Dimensions:
Usability, Playability, Narratives, Enjoyment, etc.

Measurement:
Again, a subjective scale with Likert-scale questions



Interaction Design

Allocating functions to humans and/or technology

Involves sequence/structure of touchpoints of space and time

=> It’s not just about efficiency, it's about user experience as a whole!

Benyon, D. (2019). Designing User Experience: a guide to HCI, UX and interaction design. Pearson UK

Interaction Design (IxD)

User Experience (ISO 9241-210)

- **“User’s perception and responses that result from the use and/or anticipated use of a system, product, or service”**
  - Users’ perceptions and responses include the users’ emotions, beliefs, preferences, comfort, behaviors, and accomplishments that occur before, during, and after use.
  - User experience is as consequence of brand image, presentation, functionality, system performance, interactive behavior, and assistive capabilities of a system, product, or service. It also results from the users’ internal and physical state resulting from prior experiences, attitudes, skills, abilities and personality; and from the context of use

Interaction Design (IxD)

User Experience (UX)



Human-Machine Interaction

UX Examples: MRI



Human-Machine Interaction

UX Examples: MRI



Interaction Design (IxD)

UX Examples: Children’s Hospital in Pittsburg (Doug Dietz)



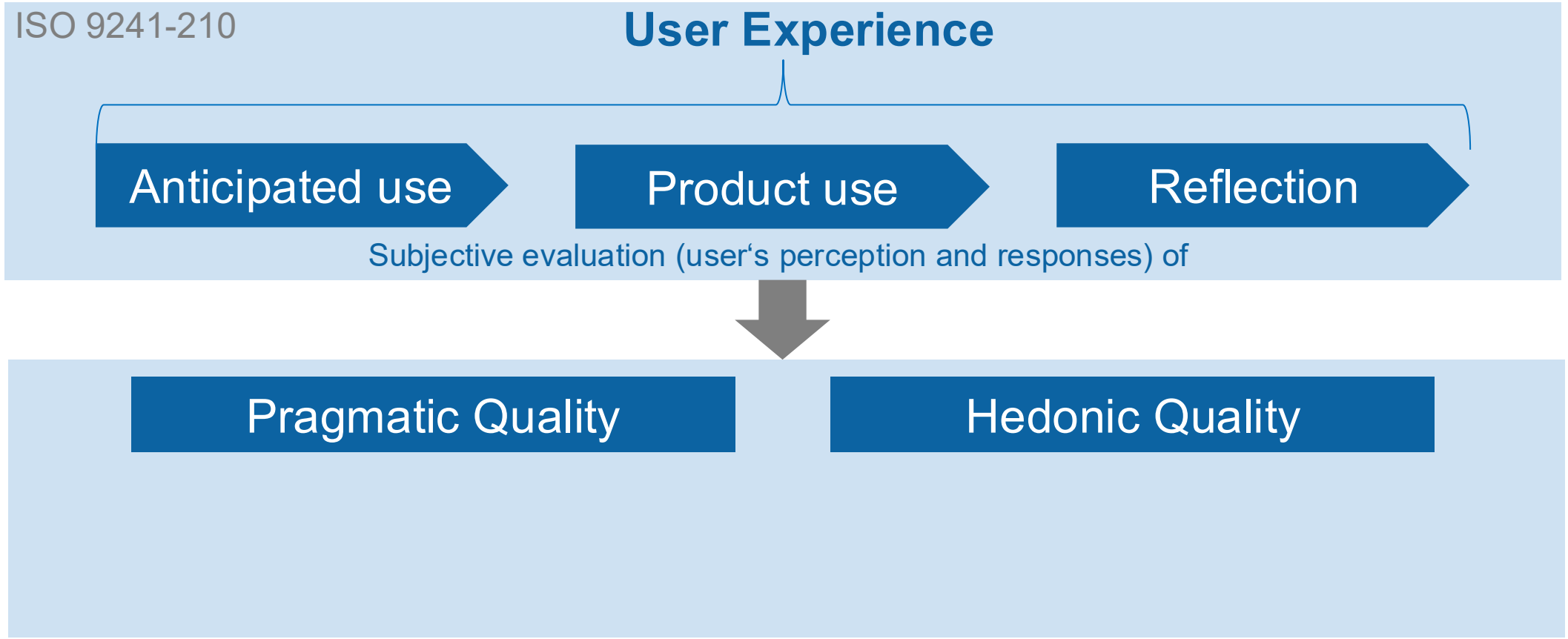
Interaction Design (IxD)

UX Examples: Children's Hospital in Pittsburg (Doug Dietz)



Interaction Design (IxD)

User Experience (UX)



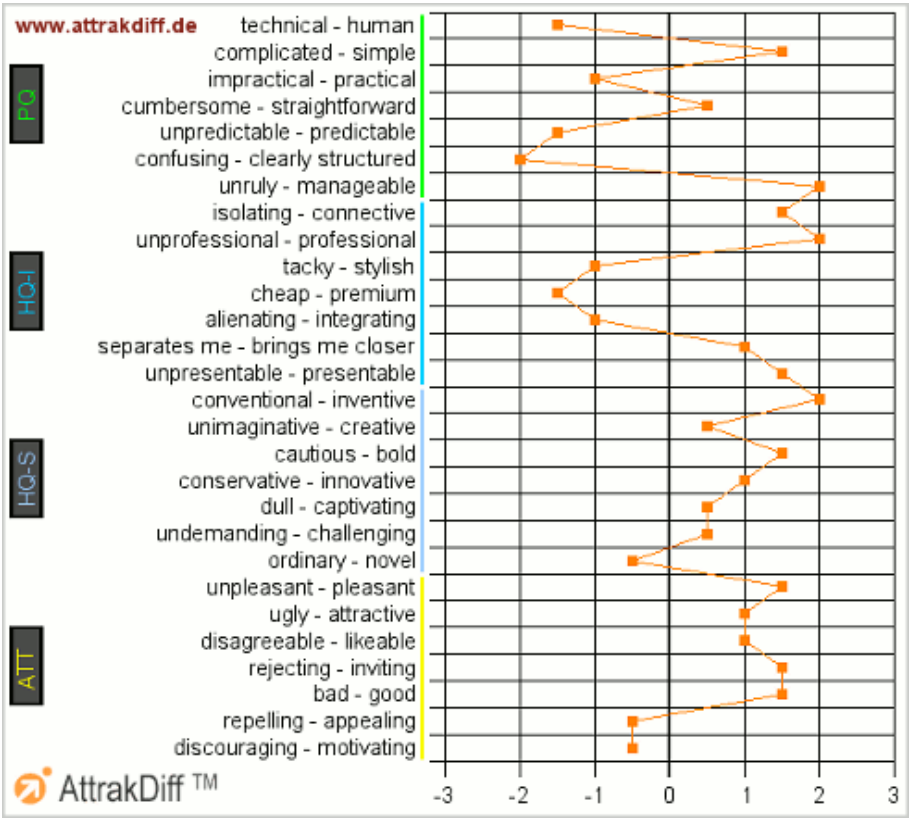
Hassenzahl, M. (2007). The hedonic/pragmatic model of user experience. Towards a UX manifesto, 10.

Interaction Design (IxD)

User Experience (UX)

- **Measurements**
  - **User Experience Questionnaire (UEQ)**
    - https://www.ueq-online.org/
    - Long (26 items) and short (8 items) version
    - Dimensions: Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, Novelty
  - **AttrakDiff**
    - http://www.attrakdiff.de/
    - 28 items
    - Dimensions: Attractiveness, Pragmatic Quality, Hedonic Qualities (HQS, HQI)

obstructive	o o o o o o o	supportive	1
complicated	o o o o o o o	easy	2
inefficient	o o o o o o o	efficient	3
clear	o o o o o o o	confusing	4
boring	o o o o o o o	exiting	5
not interesting	o o o o o o o	interesting	6
conventional	o o o o o o o	inventive	7
usual	o o o o o o o	leading edge	8



Interaction Design (IxD)

User Experience (UX)

- **UX: strong relation with emotions and psychological needs**
  - "[…] a momentary, primarily evaluative feeling (good-bad) while interacting with a product or service. By that, UX shifts attention from the product and materials (i.e., content, functions, presentation, interaction) to humans and feelings – the subjective side of product use."
- Hassenzahl (2010): **Good UX** is the consequence of **fulfilling the human needs** for
  - Autonomy
  - Competence
  - Stimulation
  - Relatedness
  - Popularity
  - Meaning
  - Security
- These needs can be assessed systematically

Hassenzahl, M., Diefenbach, S., & Göritz, A. (2010). Needs, affect, and interactive products–Facets of user experience. Interacting with computers, 22(5), 353-362.

Interaction Design (IxD)

User Experience (UX)

- Need Scale: “During this event/while using [X] I felt…”
  - **Autonomy**
    - That my choices were based on my true interests and values
    - Free to do things my own way
    - That my choices expressed my “true self”
  - **Competence**
    - That I was successfully completing difficult tasks and projects
    - That I was taking on and mastering hard challenges
    - Very capable in what I did
  - **Stimulation**
    - That I was experiencing new sensations and activities
    - Intense physical pleasure and enjoyment
    - That I found new sources and types of stimulation for myself
  - **Relatedness**
    - A sense of contact with people who care for me, and whom I care for
    - Close and connected with other people who are important to me
    - A strong sense of intimacy with the people I spent time with
  - **Popularity**
    - That I was a person whose advice other seek out and follow
    - That I strongly influenced others‘ beliefs and behavior
    - That I had strong impact on what other people did
  - **Meaning**
    - That I was “becoming who I really am”
    - A sense of deeper purpose in life
    - A deeper understanding of myself and my place in the universe
  - **Security**
    - That my life was structured and predictable
    - Glad that I have a comfortable set of routines and habits
    - Safe from threats and uncertainties

Human-Machine Interaction

UX Examples: “Hotzenplotz” Coffee Grinder





Human-Machine Interaction

UX Examples: “Hotzenplotz” Coffee Grinder



Manual Coffee Grinder:
Rich, emotionally intensive experience by the manual action combined with smell, but physically exhausting



Automated Coffee Grinder:
Single button, “out-of-the-loop”, less meaningful and enjoyable, “cost of automation”

Human-Machine Interaction

UX Examples: “Hotzenplotz” Coffee Grinder



Klapperich, H., & Hassenzahl, M. (2016, October). Hotzenplotz: Reconciling automation with experience. In Proc. of the 9th Nordic conference on human-computer interaction (pp. 1-10).

Human-Machine Interaction

UX Examples: “Hotzenplotz” Coffee Grinder

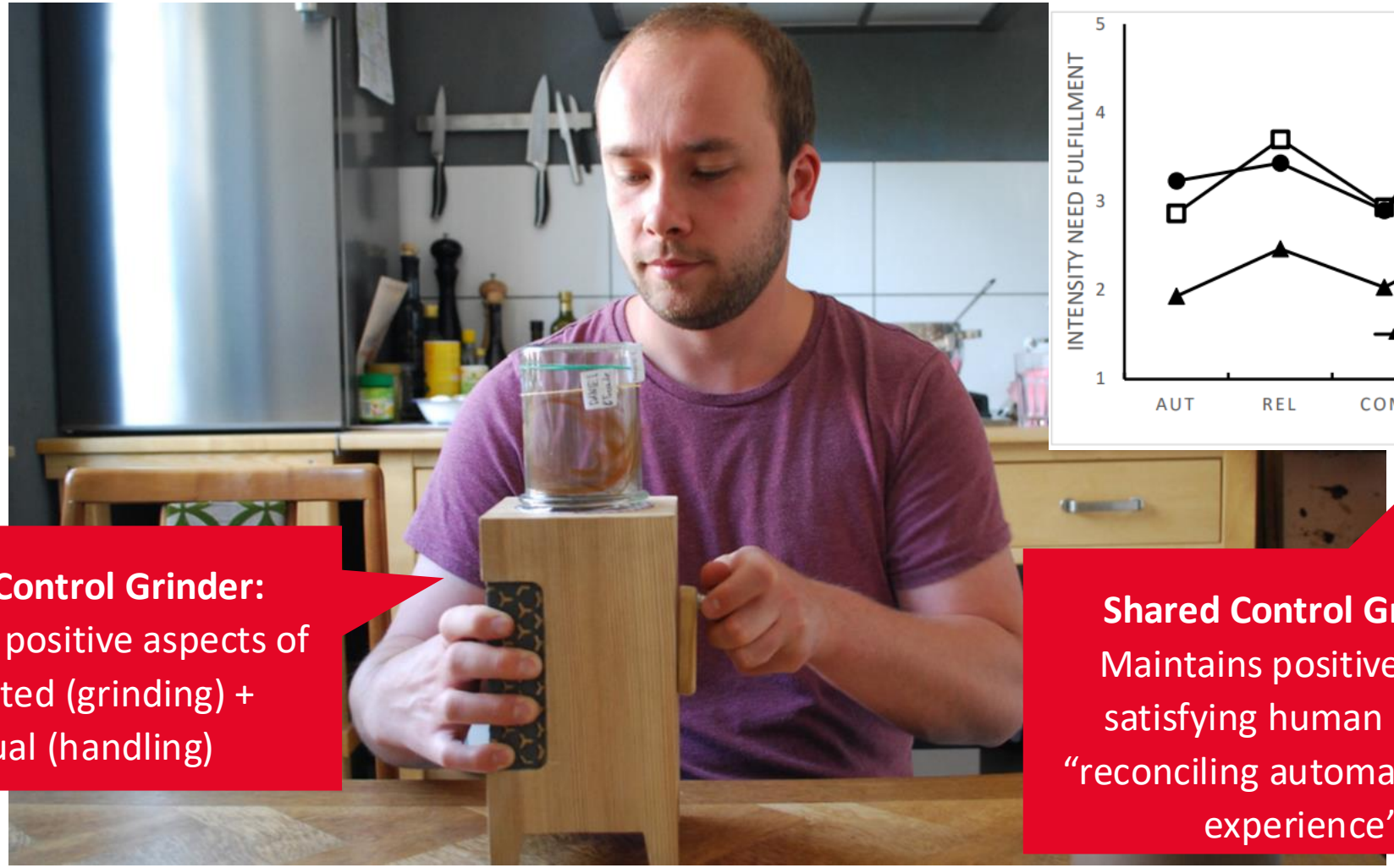


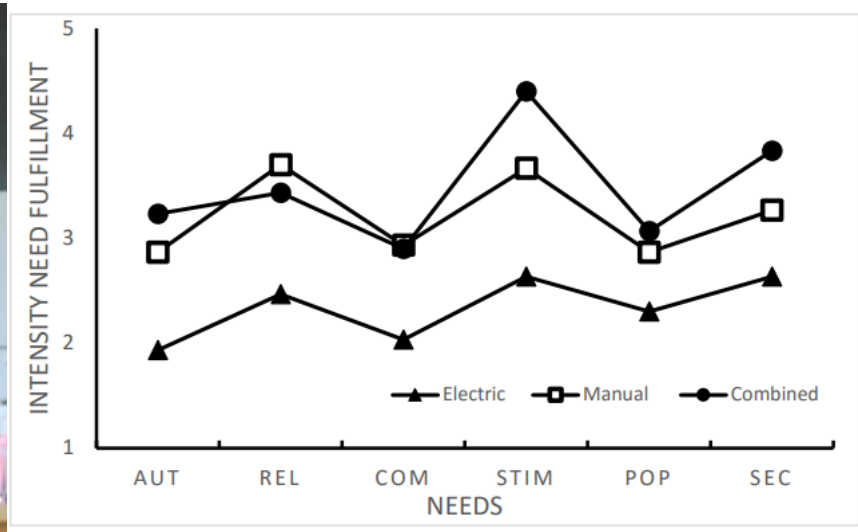
Shared Control Grinder:
Combining positive aspects of automated (grinding) + manual (handling)

Klapperich, H., & Hassenzahl, M. (2016, October). Hotzenplotz: Reconciling automation with experience. In Proc. of the 9th Nordic conference on human-computer interaction (pp. 1-10).

Human-Machine Interaction

UX Examples: “Hotzenplotz” Coffee Grinder





Shared Control Grinder:
Combining positive aspects of automated (grinding) + manual (handling)

Shared Control Grinder:
Maintains positive UX by satisfying human needs, “reconciling automation with experience”

Klapperich, H., & Hassenzahl, M. (2016, October). Hotzenplotz: Reconciling automation with experience. In Proc. of the 9th Nordic conference on human-computer interaction (pp. 1-10).

Human-Machine Interaction

UX Examples: “Driving Hotzenplotz”, Adaption to Automated Vehicles



Frison, A. K., Wintersberger, P., Riener, A., & Schartmüller, C. (2017, September). Driving Hotzenplotz: A Hybrid Interface for Vehicle Control Aiming to Maximize Pleasure in Highway Driving. In Proceedings of the 9th international conference on automotive user interfaces and interactive vehicular applications (pp. 236-244).

Human-Machine Interaction

UX Examples: “Driving Hotzenplotz”, Adaption to Automated Vehicles

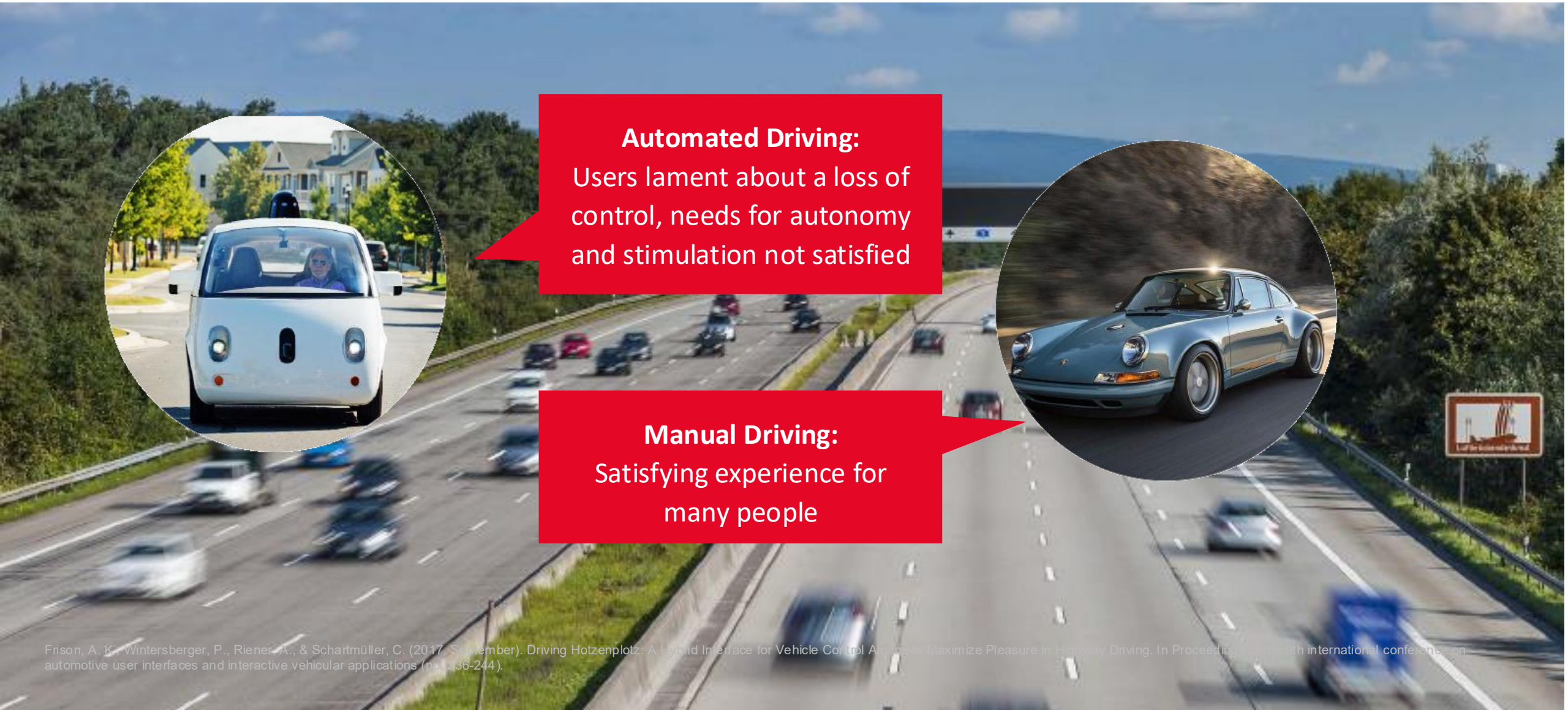


Automated Driving:
Users lament about a loss of control, needs for autonomy and stimulation not satisfied

Frison, A. K., Wintersberger, P., Riener, A., & Schartmüller, C. (2017, September). Driving Hotzenplotz: A Hybrid Interface for Vehicle Control Aiming to Maximize Pleasure in Highway Driving. In Proceedings of the 9th international conference on automotive user interfaces and interactive vehicular applications (pp. 236-244).

Human-Machine Interaction

UX Examples: “Driving Hotzenplotz”, Adaption to Automated Vehicles



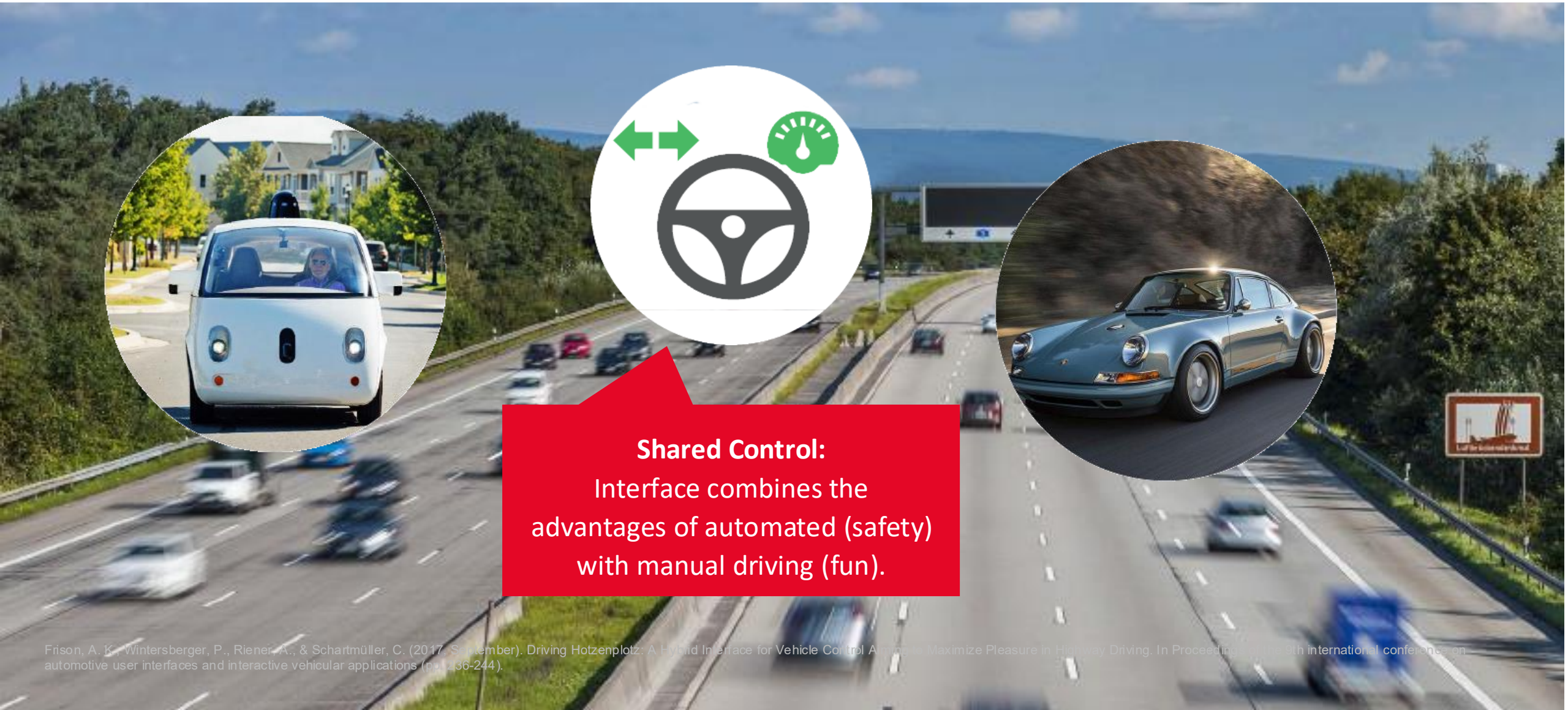
Automated Driving:
Users lament about a loss of control, needs for autonomy and stimulation not satisfied

Manual Driving:
Satisfying experience for many people

Frison, A. K., Wintersberger, P., Riener, A., & Schartmüller, C. (2017, September). Driving Hotzenplotz: A Hybrid Interface for Vehicle Control Aiming to Maximize Pleasure in Highway Driving. In Proceedings of the 9th international conference on automotive user interfaces and interactive vehicular applications (pp. 236-244).

Human-Machine Interaction

UX Examples: “Driving Hotzenplotz”, Adaption to Automated Vehicles



Shared Control:
Interface combines the advantages of automated (safety) with manual driving (fun).

Frison, A. K., Wintersberger, P., Riener, A., & Schartmüller, C. (2017, September). Driving Hotzenplotz: A Hybrid Interface for Vehicle Control Aiming to Maximize Pleasure in Highway Driving. In Proceedings of the 9th international conference on automotive user interfaces and interactive vehicular applications (pp. 236-244).



To foster the **joy of driving** we built a **hybrid interface**

Human-Machine Interaction

UX Examples: “Driving Hotzenplotz”, Adaption to Automated Vehicles



“Hotzenplotz Driving”:
Just as the coffee grinder, our interface could “reconcile automation with (user) experience”

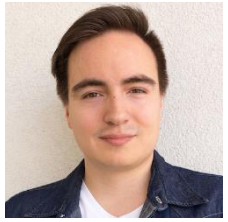
Frison, A. K., Wintersberger, P., Riener, A., & Schartmüller, C. (2017, September). Driving Hotzenplotz: A Hybrid Interface for Vehicle Control Aiming to Maximize Pleasure in Highway Driving. In Proceedings of the 9th international conference on automotive user interfaces and interactive vehicular applications (pp. 236-244).

Game Engine Architectures

FH Oberösterreich
University of Applied Sciences Upper Austria

Digital Media Department

Media Technology & Design
Digital Arts
Interactive Media



Dr. Andreas Riegler
Professor of Interactive Systems
Digital Media Department
E Andreas.Riegler@fh-hagenberg.at